Metadata sheet for “A Comparison of LSTM and Random Forest Models for Multi-Lead-Time River Discharge Forecasting” from Haberstock et al. 2026

Data file	**Metadata**	**Definition**
Einen_Ems_Precipitation_Discharge_DOY.csv	Data format	Comma Separated-Values File
	Description	Data set with the combined data of the seven rainfall stations, discharge and DOY. This data set is used for running the ML models
	Author	Tommi Haberstock
	Date	13.01.2026
	Source	Github > Haberstock-et-al.-2026-SupplementaryMaterial/ML_Input_Data/
	Origin of Data	Parts of the data have been requested State Office for Nature, Environment, and Climate North RhineWestphalia (LANUK). Parts of ther data have been downloaded from OpenGeodata.NRW - Hydrologische Daten https://www.opengeodata.nrw.de/produkte/umwelt_klima/wasser/ -oberflaechengewaesser/-hydro/
augustdorf_2007-2024.csv, bielefeld-mitte_2007-2024.csv, delbrueck-steinhorst_2007-2024.csv, rheda-wiedenbrueck_2007-2024.csv, rietberg_2007-2024.csv, Versmold-Vorbruch_2007-2025.csv, warendorf-freckenhorst_2007-2024.csv, Einen_Stunden_2007-2024.csv	Data format	Comma Separated-Values File
	Description	The seven rainfall measuring stations and the discharge gauge at Einen. All data sets were cropped to the same timespan (01.01.2007 - 01.01.2024).
	Author	Tommi Haberstock
	Date	04.04.2026
	Source	Github > Haberstock-et-al.-2026-SupplementaryMaterial/Input_Data_Stations/
	Origin of Data	Requested from the State Office for Nature, Environment, and Climate North Rhine-Westphalia (LANUK); Edited
rf_predictions_H3_test.csv, rf_predictions_H6_test.csv, rf_predictions_H12_test.csv, rf_predictions_H24_test.csv, rf_predictions_H72_test.csv,	Data format	Comma Separated-Values File
	Description	Saved predictions of the RF model for all five horizons. These files are loaded and used to skip the repeated running of the RF model
	Author	Tommi Haberstock
	Date	24.01.2026
	Source	Github > Haberstock-et-al.-2026-Supplementary-Material /rf_predictions_best_compromise/
	Origin of Data	Downloaded from the 02_Haberstock_Ems_RF.ipynb Python Notebook
lstm_h3.keras, lstm_h6.keras, lstm_h12.keras, lstm_h24.keras, lstm_h72.keras	Data format	Keras format
	Description	Saved data files of the ran LSTM model with the highest NSE values for all horizons. These files are loaded and used to skip the repeated running of the LSTM model
	Author	Tommi Haberstock
	Date	22.01.2026
	Source	Github> Haberstock-et-al.-2026-SupplementaryMaterial/trained_lstm_models/
	Origin of Data	Downloaded from the 03_Haberstock_Ems_LSTM.ipynb Python Notebook

predictions_h3_with_doy.csv predictions_h6_with_doy.csv predictions_h12_with_doy.csv predictions_h24_with_doy.csv predictions_h72_with_doy.csv	Data format	Comma Separated-Values File
	Description	Saved .csv files of the ran LSTM model containing the observed and predicted values of the test data set.
	Author	Tommi Haberstock
	Date	25.03.2026
	Source	Github > Haberstock-et-al.-2026-SupplementaryMaterial/predictions_with_doy/
	Origin of Data	Downloaded from the 03_Haberstock_Ems_LSTM.ipynb Python Notebook
01_Haberstock_Ems_Data_Processing.ipynb	Data format	Interactive Python Notebook
	Description	This Jupyter Notebook is the first of this study. Here, the seven precipitation stations, the discharge data and DOY are combined to one data set. Furthermore, a IDW interpolation i performed and a cross correlation between discharge and rainfall created. Together with the autocorrelation of the river's discharge, a performance model is created for model evaluation.
	Author	Tommi Haberstock
	Date	05.08.2026
	Source	Github > Haberstock-et-al.-2026-SupplementaryMaterial/Google Colab Jupyter Notebooks/
	Language	English
02_Haberstock_Ems_RF.ipynb	Data format	Interactive Python Notebook
	Description	This is the second Jupyter Notebook. Here, the Random Forest model was created and evaluated.
	Author	Tommi Haberstock
	Date	05.08.2026
	Source	Github > Haberstock-et-al.-2026-SupplementaryMaterial/Google Colab Jupyter Notebooks/
	Language	English
03_Haberstock_Ems_LSTM.ipynb	Data format	Interactive Python Notebook
	Description	In the third Jupyter Notebook, the LSTM model was created and evaluated.
	Author	Tommi Haberstock
	Date	05.08.2026
	Source	Github > Haberstock-et-al.-2026-SupplementaryMaterial/Google Colab Jupyter Notebooks/
	Language	English
04_Haberstock_Ems_Model_Comparison.ipynb	Data format	Interactive Python Notebook
	Description	In the fourth and last Jupyter Notebook, the two created ML models were compared to each other and graphs created to visualise the difference in performance.
	Author	Tommi Haberstock
	Date	05.08.2026
	Source	Github > Haberstock-et-al.-2026-SupplementaryMaterial/Google Colab Jupyter Notebooks/
	Language	English